MP352
Special Relativity

Time allowed: $1\frac{1}{2}$ hours

Answer **ALL** questions

This is a **SAMPLE** exam, roughly reflecting the general structure of the finals for 2016 – 2017.

Remember that **ALL** questions are to be answered.

1. Let Σ and Σ' be inertial frames. Frame Σ' moves at velocity v with respect to Σ, in the common (positive) x direction. Measurements of events in the two frames, denoted respectively by (x, y, z, t) and (x', y', z', t'), are related by the Lorentz transformation (LT)

$$x' = \gamma_v(x - vt) \ ; \quad y' = y \ ; \quad z' = z \ ; \quad t' = \gamma_v(t - vx/c^2)$$

where $\gamma_v = (1 - v^2/c^2)^{-1/2}$.

(a) A photon has velocity $\vec{u}' = (\frac{2}{3}c, \frac{2}{3}c, \frac{1}{3}c)$ relative to Σ'.
Find the velocity of the photon relative to Σ.
Explain how your result is consistent with the constancy of the speed of light.

[23 marks]

(b) Represent the (ct, x) and the (ct', x') axes on a single spacetime diagram, such that the ct and x axes are perpendicular to each other.
Use the LT to find out how x' units are related to x units on this diagram.
(Hint: You could consider the event $(ct', x') = (0, 1)$, find its coordinates in the Σ frame, and hence obtain the distance of this point from the origin in x units.)

[15 marks]

(c) A particle moves with speed $(u, 0, 0)$ relative to Σ. Write down the four-velocity of the paricle as observed from the Σ frame and as observed from the Σ' frame.

[12 marks]

2. (a) Lorentz boosts in the x direction can be represented be a transformation matrix of the form

$$\begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix}$$

where the y and z coordinates have been ignored.

Show that the set of all boosts in the x direction form a group.

Will the set of all boosts in ALL directions also form a group? Explain why, or why not.

[20 marks]

(b) A neutral pion of rest mass m and (relativistic) momentum $p = \frac{3}{4}mc$ decays into two photons, One of the photons is emitted in the same direction as the original pion, and the other in the opposite direction. Find the energy of each photon.

[15 marks]

(c) Measured in one intertial frame, events A and B have spatial coordinates

$$(x_A,\, y_A,\, z_A) \;=\; (4L,\, -6L,\, 0)\ , \qquad (x_B,\, y_B,\, z_B) \;=\; (7L,\, -2L,\, 0)\ ,$$

and temporal coordinates

$$t_A \;=\; 2L/c\ , \qquad t_B \;=\; 12L/c\ ,$$

where L is a positive constant.

Calculate the invariant interval between the events. Is this interval timelike, spacelike, or null?

Explain whether there exists a different inertial frame in which the two events occur simultaneously.

[15 marks]